Domain Generalization in Abstractive Summarization:
Fine-Tuning T5 on Biomedical Text and Evaluating Cross-Domain Transfer

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https://github.com/jeremy-j-liu/NLP-Project/blob/main/NLP_ProjectOutput.ipynb

Abstract

This project investigates the domain generalization capability of a fine-tuned T5-small model for abstractive summarization of scientific literature. We fine-tune T5-small on the PubMed biomedical summarization dataset and evaluate performance against a Lead-3 extractive baseline and a zero-shot BART-large-cnn model. We further apply the fine-tuned model to arXiv papers without retraining, quantifying the domain transfer gap (biomedical →computer science/general science). A truncation ablation compares Lead-512 and Tail-512 input strategies, testing whether biomedical IMRaD structure favors tail truncation. Contrary to this hypothesis, Lead-512 outperforms Tail-512 across all ROUGE metrics. The fine-tuned T5 model achieves ROUGE-1 of 0.286 on PubMed and 0.301 on arXiv, with near-identical BERTScore F1 (0.7803 vs. 0.7802), indicating strong cross-domain generalization. Error analysis identifies multilingual data contamination, article length, and repetitive degeneration as the primary sources of poor performance.

1. Introduction

The volume of published biomedical and scientific literature has grown at an unprecedented rate, making it increasingly difficult for researchers and clinicians to stay current. Automatic summarization systems offer a promising solution by condensing lengthy research articles into concise, informative abstracts. However, the specialized vocabulary, structured argumentation (IMRaD: Introduction, Methods, Results, Discussion), and high factual precision requirements of scientific text pose unique challenges that general-purpose summarization models are not designed to address.

This work addresses two central research questions. First, does fine-tuning a general-purpose sequence-to-sequence model on biomedical text produce measurable improvements over extractive and zero-shot abstractive baselines? Second, does biomedical fine-tuning generalize to other scientific domains, or does it introduce domain-specific overfitting that degrades out-of-distribution performance? We further examine whether the structural properties of PubMed articles, specifically the placement of key findings relative to article position, influence optimal input truncation strategy, a methodological choice that has received limited systematic study in the biomedical NLP literature.

2. Methods

2.1 Data

We use two datasets. For in-domain training and evaluation we use ccdv/pubmed-summarization, comprising over 200,000 PubMed article-abstract pairs. We train on 5,000 samples and evaluate on a held-out test slice of 100 samples. For cross-domain evaluation we use ccdv/arxiv-summarization on the same 100-sample test slice without retraining. All text is preprocessed to remove inline LaTeX math, LaTeX commands, and numeric citation brackets prior to tokenization.

2.2 Models and Baselines

Three systems are evaluated. **(1) Lead-3:** an extractive baseline returning the first three sentences, establishing a lower bound. **(2) BART-large-cnn (zero-shot):** the pretrained BART model applied without fine-tuning, using a hybrid head+tail strategy concatenating the first 511 and last 510 tokens separated by an EOS token. **(3) T5-small (fine-tuned):** T5-small fine-tuned on PubMed using the HuggingFace Seq2SeqTrainer with a summarize: task prefix, trained for 3 epochs, batch size 8, gradient accumulation 2, learning rate 5e-4, fp16 precision on a T4 GPU.

2.3 Truncation Ablation

To test whether IMRaD structure favors tail truncation, we compare Lead-512 (first ~600 words) against Tail-512 (last ~600 words) using the fine-tuned T5 model on 100 PubMed test samples, directly examining whether lead bias transfers from news to biomedical literature.

2.4 Evaluation Metrics

We report ROUGE-1, ROUGE-2, ROUGE-L, and ROUGE-Lsum for n-gram overlap. BERTScore F1, Precision, and Recall using distilbert-base-uncased are reported for semantic similarity. For error analysis we compute per-sample ROUGE-1, article length vs. ROUGE-1 Pearson correlation, out-of-vocabulary (OOV) rates relative to the PubMed test vocabulary, and sentence-level repetition rates across prediction sets.

3. Results and Discussion

3.1 Phase 1 Baselines: Lead-3 vs. BART-Hybrid (Zero-Shot)

Metric	Lead-3	BART-Hybrid
ROUGE-1	0.2658	0.2485
ROUGE-2	0.0908	0.0922
ROUGE-L	0.1697	0.1642
ROUGE-Lsum	0.2220	0.2088

Table 1: Phase 1 baseline comparison on PubMed test set (100 samples). Shaded cell = BART-Hybrid beats Lead-3.

Prior to fine-tuning, we establish two baselines on the PubMed test slice (Table 1). Lead-3 outperforms BART-Hybrid on three of four ROUGE metrics, with BART-Hybrid achieving a marginal advantage only on ROUGE-2 (0.0922 vs. 0.0908). This confirms that zero-shot BART with hybrid head+tail truncation does not reliably exceed a simple extractive baseline on biomedical text, motivating domain-specific fine-tuning.

3.2 In-Domain Performance After Fine-Tuning (PubMed)

Metric	Lead-3	BART-0shot	T5-FT (ours)
ROUGE-1	0.2655	0.1669	0.2860
ROUGE-2	0.0917	0.0337	0.1064
ROUGE-L	0.1700	0.1047	0.1865
ROUGE-Lsum	0.2232	0.1386	0.2419
BERTScore F1	0.7717	0.7281	0.7803

Table 2: Full head-to-head comparison on PubMed test set (100 samples). Shaded column = best per metric.

Fine-tuned T5 outperforms both baselines across all metrics (Table 2). The ROUGE-2 improvement from Lead-3 to T5-FT (0.092 to 0.106) is particularly meaningful as ROUGE-2 is the most discriminating metric, requiring exact bigram matches. Zero-shot BART performs worse than Lead-3 across all ROUGE metrics

(ROUGE-1: 0.167 vs. 0.266), consistent with domain adaptation literature: BART pre-trained on news text encounters significant vocabulary and structural distribution shift on dense biomedical prose. The BERTScore gap between Lead-3 and T5-FT is narrower (0.772 vs. 0.780), suggesting fine-tuning primarily improves lexical precision rather than topical coverage.

3.3 Truncation Ablation

Strategy	ROUGE-1	ROUGE-2	ROUGE-L	Winner
Lead-512	0.2859	0.1052	0.1867	Lead
Tail-512	0.2501	0.0569	0.1485	Lead
Delta	+0.0358	+0.0483	+0.0382	Lead

Table 3: Lead-512 vs. Tail-512 truncation ablation (T5-FT, PubMed). Shaded row = delta.

The truncation ablation yields a counterintuitive result: Lead-512 outperforms Tail-512 by a substantial margin across all metrics, with the ROUGE-2 delta of +0.048 being particularly striking (Table 3). This contradicts the hypothesis that IMRaD-structured text would favor tail truncation. We interpret this as evidence that PubMed abstracts are structured overviews drawing on background, methods, and results — content distributed throughout the article. The lead bias documented in news summarization appears to extend to structured scientific writing, though for different structural reasons.

3.4 Cross-Domain Transfer (arXiv)

Metric	PubMed (in)	arXiv (out)	Delta (out−in)
ROUGE-1	0.2860	0.3006	+0.0146
ROUGE-2	0.1064	0.0850	−0.0214
ROUGE-L	0.1865	0.1790	−0.0075
BERTScore F1	0.7803	0.7802	−0.0001

Table 4: Domain gap analysis — T5-FT on PubMed vs. arXiv. Positive delta = improvement on arXiv.

The cross-domain evaluation reveals surprising robustness (Table 3). ROUGE-1 is marginally higher on arXiv than PubMed, and BERTScore F1 is effectively identical (0.7803 vs. 0.7802). The only meaningful degradation occurs in ROUGE-2 (drop of 0.021), suggesting the model retains topical coherence across domains but loses domain-specific phrase-level fluency. A key caveat is that the arXiv sample drew from physics and astrophysics papers rather than only computer science, which shares closer structural properties with PubMed than computer science papers would. Despite an OOV rate of 15.7% on arXiv relative to PubMed vocabulary, the OOV-ROUGE-1 correlation was only r = -0.20, indicating vocabulary mismatch alone is a modest predictor of performance.

3.5 Score Distributions and Length Effects

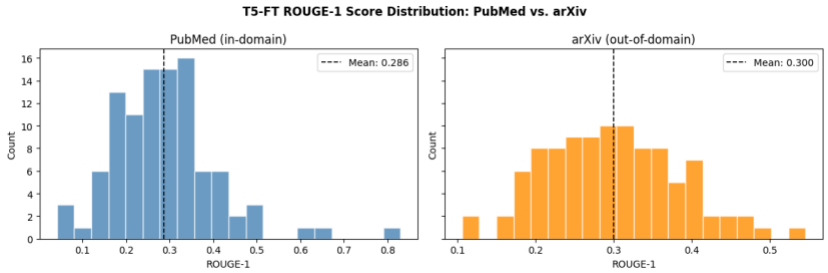

Figure 1: T5-FT ROUGE-1 score distributions for PubMed and arXiv test sets.

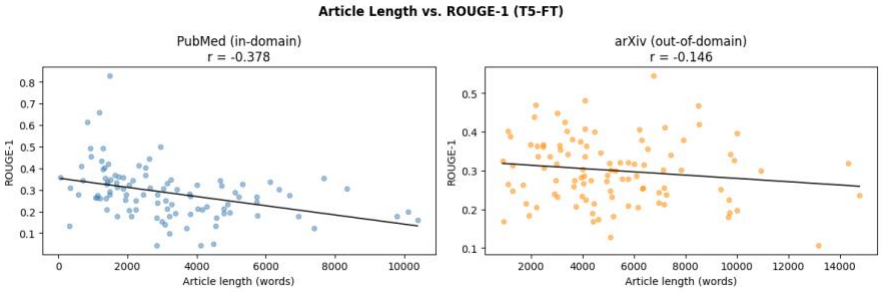

Figure 2: Article length vs. ROUGE-1 (T5-FT). Pearson r = -0.378 (PubMed), r = -0.146 (arXiv).

Figure 1 shows the PubMed distribution has a heavier left tail — four samples score below 0.10 versus zero on arXiv — indicating catastrophic failures are more prevalent in-domain. Investigation reveals these are primarily multilingual articles (Polish and Portuguese), a data quality issue rather than a model limitation. Figure 2 confirms a meaningful negative correlation between article length and ROUGE-1 on PubMed (r = -0.378), consistent with the 512-token input bottleneck being more exposed in longer articles. The weaker arXiv correlation (r = -0.146) may reflect the more uniform length distribution of physics preprints.

3.6 Error Analysis

Manual review of the bottom-10 PubMed predictions reveals three primary failure categories. **Multilingual contamination** (Samples 66, 5): the model produces output in Polish and Portuguese, as PubMed's international indexing introduces non-English articles not filtered during preprocessing. These two samples account for the most extreme ROUGE-1 failures (0.042, 0.044). **Method-level hallucination** (Samples 8, 60, 89): the model fixates on specific procedural details such as RNA extraction protocols, buffer solution compositions, sequencing library preparation, producing technically plausible but topically wrong summaries describing methods rather than findings. **Background drift** (Samples 67, 90, 7): summaries correctly identify the disease or topic but reproduce introductory background context instead of the study's specific contribution.

Sentence-level repetition, a known T5-small degeneration failure mode, was detected in 7% of PubMed predictions and 6% of arXiv predictions, suggesting this as a model capacity issue independent of domain. Zero-shot BART produced incoherent, garbled output across all qualitative examples in both domains, with the hybrid head+tail concatenation appearing to disrupt BART's positional encoding expectations, suggesting naive boundary concatenation is incompatible with models pre-trained on contiguous text.

4. Conclusion

This project presents a systematic study of abstractive summarization for scientific literature, examining fine-tuning effectiveness, truncation strategy, and cross-domain generalization. Fine-tuned T5-small substantially outperforms both the Lead-3 extractive baseline and zero-shot BART on PubMed, confirming that domain-specific fine-tuning is warranted. The truncation ablation produced a notable null result: Lead-512 outperforms Tail-512 across all metrics, suggesting the lead bias from news summarization extends to structured scientific text contrary to the IMRaD-motivated hypothesis. Cross-domain evaluation on arXiv reveals strong generalization with near-zero BERTScore degradation, though the physics-heavy arXiv sample warrants caution in generalizing this to computer science papers.

Error analysis identifies multilingual data contamination as a critical preprocessing gap that disproportionately inflates failure rates, language detection filtering should be a suggested step for PubMed datasets. Future work should evaluate longer-context models such as Longformer or LED to address the input truncation bottleneck identified by the strong length-ROUGE correlation. Factual consistency

evaluation beyond ROUGE, particularly negation-aware metrics for medical text, remains an important open problem.

View project code below:
https://github.com/jeremy-j-liu/NLP-Project/blob/main/NLP_ProjectOutput.ipynb